

Triangular truncation on nuclear ideals: an exact reverse-exponent classification

Strong partial answer to Problem 7.4 of arXiv:1501.03267

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Status: strong partial result; likely valid; human review needed. For ℓ_∞ interpreted as c_0 , triangular truncation is bounded on $\mathcal{N}(\ell_a, \ell_b)$ exactly when $b < a$ or $a = b \in \{1, \infty\}$. Outside this region its finite-section norms grow at least logarithmically.

Source problem

Rozendaal, Sukochev, and Tomskova ask for which sequence-space exponents the standard triangular truncation is bounded on (p, q) -summing ideals, and whether other non-trivial operator ideals have bounded triangular truncation [1, Problem 7.4, p. 25]. The exact source text is reproduced below.

Problem 7.4. Let $p, q, r \in [1, \infty]$ be such that $q \geq 2$ and $\frac{1}{q} - \frac{1}{p} < \frac{1}{2}$ and $\frac{1}{r} = \frac{1}{p} - \frac{1}{q} + \frac{1}{2}$. Then the Schatten class $\mathcal{S}_r$ coincides with the Banach ideal $\Pi_{p,q}(\ell_2)$ of (p, q) -summing operators on ℓ_2 (for the definition of (p, q) -summing operators see [13]). Hence, by [14], the standard triangular truncation is bounded on $\Pi_{p,q}(\ell_2)$ for these indices. For which $r_1, r_2 \in [1, \infty]$ is the standard triangular truncation bounded on $\Pi_{p,q}(\ell_{r_1}, \ell_{r_2})$? Are there other non-trivial ideals $\mathcal{I}$ in $\mathcal{L}(\ell_p, \ell_q)$ such that the standard triangular truncation is bounded on $\mathcal{I}$? As shown in Theorem 7.3 (and the rest of this paper), answers to these questions are linked to commutator estimates for diagonalizable operators.

This packet completely classifies one classical ideal, the nuclear ideal, and thereby answers the source's existence clause affirmatively. It does not classify the requested (p, q) -summing ideals.

Theorem and proof intuition

For $1 \leq a \leq \infty$, write $E_a = \ell_a$ when $a < \infty$ and $E_\infty = c_0$. If $T = (t_{ij})$, let

$$(\Delta T)_{ij} = \begin{cases} t_{ij}, & i \leq j, \\ 0, & i > j. \end{cases}$$

Theorem 1. Let $1 \leq a, b \leq \infty$.

1. The standard triangular truncation extends to a bounded operator on

$$\mathcal{N}(E_a, E_b)$$

if and only if

$$b < a, \quad \text{or} \quad a = b = 1, \quad \text{or} \quad a = b = \infty.$$

2. In the two equal-endpoint cases its nuclear-ideal norm is exactly one.
3. In every remaining case there is $c_{a,b} > 0$ such that the restriction Δ_n to the first n coordinates satisfies

$$\|\Delta_n\|_{\mathcal{L}(\mathcal{N}(\ell_a^n, \ell_b^n))} \geq c_{a,b} \log n \quad (n \geq 2).$$

Proof intuition. The nuclear norm is trace-dual to the ordinary operator norm with the arrow reversed. Under this duality, upper triangular truncation becomes lower triangular truncation. Reversing both coordinate orders turns lower back into upper, so the finite nuclear truncation norm from ℓ_a^n to ℓ_b^n is *exactly* the ordinary truncation norm from ℓ_b^n to ℓ_a^n . The known full-operator phase diagram can therefore be read backwards. Only ℓ_1 and c_0 require separate endpoint arguments, and there rank-one tensors admit explicit contractive triangular decompositions.

Finite-dimensional trace duality

Put $E = \ell_a^n$ and $F = \ell_b^n$. The standard isometric duality

$$\mathcal{N}(E, F)^* = \mathcal{L}(F, E), \quad \langle T, S \rangle = \text{tr}(ST),$$

holds because E and F are finite-dimensional. Let Λ_n denote lower triangular truncation on $\mathcal{L}(F, E)$.

Lemma 1. *For every n ,*

$$\|\Delta_n\|_{\mathcal{L}(\mathcal{N}(\ell_a^n, \ell_b^n))} = \|\Delta_n\|_{\mathcal{L}(\mathcal{L}(\ell_b^n, \ell_a^n))}.$$

Proof. Writing $T = (t_{ij})$ and $S = (s_{ji})$ with the indices arranged according to their arrows,

$$\text{tr}(S\Delta_n T) = \sum_{i \leq j} s_{ji} t_{ij} = \text{tr}((\Lambda_n S)T).$$

Thus the trace adjoint of Δ_n is Λ_n , and their norms are equal. The coordinate-reversal isometries on ℓ_b^n and ℓ_a^n conjugate Λ_n to Δ_n , proving the displayed identity. $\square$

Passage to infinite sequence spaces

The full-operator classification quoted as Proposition 6.3 in the source [1, p. 19] says that triangular truncation on $\mathcal{L}(E_u, E_v)$ is bounded when $u < v$, or when $u = v = 1$, or when $u = v = \infty$; in all other cases its n th finite-section norm is at least a positive multiple of $\log n$. Applying this with $(u, v) = (b, a)$ and using Lemma 1 gives both the asserted finite-section lower bounds and uniform finite-section bounds when $b < a$.

If $b < a$, then necessarily $a > 1$. Finitely supported vectors are dense in E_a^* (also when $a = \infty$, since $E_\infty^* = \ell_1$) and in E_b . Consequently coordinate-finite rank-one tensors are dense in $E_a^* \widehat{\otimes}_\pi E_b = \mathcal{N}(E_a, E_b)$. The uniformly bounded, consistent finite triangular maps therefore extend uniquely to the whole nuclear ideal.

It remains to establish the two positive endpoints, where coordinate-finite functionals need not supply the preceding density argument. For a rank-one operator $T = x^* \otimes y$ on ℓ_1 , write $x^* = (\alpha_j) \in \ell_\infty$ and $y = (y_i) \in \ell_1$. With $x_i^* = (\alpha_j \mathbf{1}_{\{j \geq i\}})_j$,

$$\Delta T = \sum_{i \geq 1} x_i^* \otimes (y_i e_i), \quad \nu(\Delta T) \leq \sum_i \|x_i^*\|_\infty |y_i| \leq \|x^*\|_\infty \|y\|_1.$$

For $T = x^* \otimes y$ on c_0 , write $x^* = (\alpha_j) \in \ell_1$ and let P_j project onto the first j coordinates. Then

$$\Delta T = \sum_{j \geq 1} \alpha_j e_j^* \otimes P_j y, \quad \nu(\Delta T) \leq \sum_j |\alpha_j| \|P_j y\|_\infty \leq \|x^*\|_1 \|y\|_\infty.$$

Summing over nuclear representations and taking the defining infimum shows that Δ is contractive on both endpoint nuclear ideals. A single matrix unit gives norm at least one.

Finally, in every excluded parameter regime, a bounded infinite-dimensional map would restrict to uniformly bounded maps on all coordinate n -dimensional nuclear ideals. This contradicts the logarithmic lower bound obtained above and completes the proof of Theorem 1. $\square$

Scope and upgrade record

The theorem gives a proper non-trivial ideal with bounded truncation whenever $b < a$, precisely the reverse of the main non-endpoint positive region $a < b$ for the full operator space. Five focused routes were examined toward the harder (p, q) -summing classification: integral-ideal duality, Pietsch factorization, interpolation of the Hilbert and entrywise cases, transfer of ambient logarithmic examples, and the successful nuclear trace-duality route. The first four lose the fixed coordinate order or incur dimension-dependent norm distortion; details are recorded in the attempt note.

Bounded searches through 11 August 2026 for the exact problem and triangular truncation on summing/nuclear ideals found the source paper but no explicit later statement of this classification. Since the proof is a short consequence of established triangular-projection theory, novelty is provisional and likely modest.

References

- [1] J. Rozendaal, F. Sukochev, and A. Tomskova, *Operator Lipschitz functions on Banach spaces*, *Studia Math.* **232** (2016), 57–92; arXiv:1501.03267.